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# 21st-century skills in vocal pedagogy: a scoping review

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**Introduction:** This study examines how 21st-century skills are conceptualized and applied in vocal pedagogy in response to increasing demands for reflective, technology-mediated, and culturally responsive approaches in music education.

**Methods:** A scoping review was conducted following Arksey and O'Malley's framework and the Joanna Briggs Institute methodology. Literature published between 2014 and 2025 was systematically mapped from Scopus and EBSCOhost databases. Using PRISMA-based screening, nine studies met the inclusion criteria. Data were charted and thematically analyzed to identify conceptual definitions, pedagogical approaches, and implementation challenges.

**Results:** Findings reveal a shift from tradition-bound, repertoire-driven instruction toward more reflective, student-centered, and digitally supported learning environments. Integration of 21st-century competencies—such as critical thinking, creativity, collaboration, communication, and cultural literacy—occurs through strategies including AI-assisted feedback, microlearning, and cross-cultural pedagogical practices. However, challenges remain, including curriculum rigidity, insufficient teacher preparation, and limited professional development.

**Discussion:** The study proposes a conceptual framework positioning cognitive, creative, technological, emotional–social, and cultural dimensions as interdependent foundations of contemporary vocal education. This model underscores the need to cultivate reflective and adaptive learners capable of navigating complex artistic, digital, and intercultural contexts, with implications for curriculum innovation, teacher education, and future research.

### KEYWORDS

21st-century skills, AI-assisted learning, cultural literacy, music education, reflective teaching, scoping review, vocal pedagogy

## Introduction

The transformation of education in the 21st century has significantly reshaped teaching and learning across all disciplines, including music education. Global educational reform emphasizes developing future-ready learners equipped with cognitive, social, and technological competencies to navigate complex, interdisciplinary challenges. Within this context, vocal pedagogy, traditionally rooted in the master-apprentice model, faces growing demands to incorporate 21st-century skills that foster creativity, collaboration, critical thinking, and communication. This model, while historically central to professional voice training, has been increasingly critiqued for privileging technical “correctness,” hierarchical authority, and imitation at the expense of learner autonomy, reflective agency, and contextual adaptability (Lă, 2017; Moor, 2018; Long et al., 2014). Empirical and critical scholarship further suggests that such pedagogical structures may inadvertently constrain artistic authenticity and reinforce

inequitable power relations within conservatoire and studio-based learning environments (Colahan, 2022; Gibson, 2026).

These skills not only support artistic development but also empower students to become adaptive, reflective, and innovative vocal practitioners in a rapidly evolving world. Emerging research also highlights the transformative potential of Artificial Intelligence (AI) in music and vocal education. A bibliometric and scoping review by Mazlan et al. (2025a, b) identifies key trends and challenges in AI-driven pedagogy, demonstrating how machine learning and data analytics can enhance music learning through personalized feedback, adaptive assessment, and automated skill tracking. The study underscores that AI not only improves operational efficiency but also enriches reflective learning and music literacy, signaling a major pedagogical shift toward evidence-based and data-informed teaching practices. Within the broader framework of 21st-century skills, AI integration strengthens learners' technological fluency, critical engagement, and creative autonomy, key competencies for future-ready vocal practitioners.

Importantly, as many vocal educators operate within teacher education programs and music education departments, examining how 21st-century competencies are embedded in vocal pedagogy has direct implications for music teacher preparation, curriculum design, and professional development.

## Defining 21st-century skills

The concept of 21st-century skills encompasses a comprehensive set of competencies essential for thriving in contemporary society. These skills are often classified into cognitive, literacy, life, and social domains, including creativity, critical thinking, collaboration, communication, global awareness, and technological literacy (Bhatnagar, 2020). As Yoo (2022) notes, these skills are interdependent and reinforce holistic development in learners, enabling them to connect artistic practice with broader societal and digital contexts. Naixue et al. (2024) further emphasize that the “4Cs” framework: critical thinking, creativity, communication, and collaboration, forms the cornerstone of 21st-century learning, particularly in music contexts where teamwork, expression, and problem-solving are integral to the educational process. Recent syntheses of modern learning and microlearning further substantiate the call for pedagogical agility in music and vocal education. A dual bibliometric/scoping review of modern learning highlights the accelerating emphasis on AI-driven adaptive systems, competency-based curricula, and ethical considerations in technology-mediated instruction (Mazlan et al., 2025a, b). Complementing this, a scoping review on *bite-sized learning* demonstrates that microlearning formats, short, focused learning units delivered via flexible platforms are gaining traction across disciplines and show promising alignment with 21st-century traits such as personalization, reduced cognitive load, and learner engagement; the review specifically suggests that music educators could exploit platforms like TikTok for short-form pedagogy while tailoring content to learner interests (Mazlan et al., 2023).

Conversely, research in vocal health (Xiao and Mazlan, 2024) cautions that rapid digitalization and remote/short-form modalities have implications for vocal wellbeing and pedagogical content: vocal health requires intentional curricular inclusion of anatomy, hygiene, and prevention strategies, especially in conditions exacerbated by pandemic-era practices and extended screen-mediated

teaching. Finally, pedagogical innovation at the intercultural level such as applying reharmonization techniques from jazz to Malay Asli music, demonstrates how hybrid, cross-idiomatic instructional strategies can foster creativity and inclusivity while challenging Eurocentric norms in vocal and instrumental curricula (Mazlan et al., 2024). Together, these developments reinforce the necessity of integrating microlearning and technological affordances into vocal pedagogy but only when mediated by reflective practice and explicit attention to vocal health and cultural authenticity.

## Application in vocal and singing education

In vocal pedagogy, 21st-century skills manifest through a diverse range of pedagogical innovations. One notable area of integration involves technology-enhanced learning, where digital platforms, music production tools, and virtual performance environments expand the possibilities of creative expression and collaboration. Laato et al. (2020) highlight how technology-assisted composition and performance can strengthen both musical and computational thinking, allowing students to engage with modularity, loops, and design processes reflective of 21st-century cognitive skills. Similarly, studies on choir singing in Chinese universities reveal that choral instruction increasingly aims to cultivate the 4C competencies, fostering collaboration and communication while nurturing aesthetic sensitivity (Naixue et al., 2024).

At the professional level, vocal training programs in Europe and China have evolved toward a more holistic framework that blends technical proficiency with cultural awareness and ethical responsibility (Wu, 2025). This approach aligns with global educational trends emphasizing lifelong learning and intercultural competence. Likewise, musical theatre education has expanded beyond performance mastery to encompass adjacent creative, technological, and wellness skills, preparing students for diverse professional realities that include entrepreneurship, digital media, and mental well-being (O'Bryan and Harrison, 2024). Collectively, these examples demonstrate how vocal education in the 21st century increasingly integrates artistic, technological, and interpersonal dimensions to prepare students for complex, multifaceted careers.

## Challenges and gaps in implementation

Despite growing advocacy for integrating 21st-century skills into music and vocal education, implementation remains inconsistent and fragmented. Research indicates that many educators lack a clear understanding of how to effectively teach competencies such as creativity, collaboration, and critical thinking within the framework of vocal pedagogy (Nazarian et al., 2024; Irfiana et al., 2023). A persistent gap in professional development also limits teachers' ability to translate policy-level expectations into classroom practice (Pharamela and Singh-Pillay, 2025). Furthermore, curriculum rigidity in many institutions prevents the integration of flexible, cross-disciplinary methods aligned with 21st-century learning (Alharbi, 2024).

Additional barriers include inadequate access to technological resources, uneven digital literacy among educators, and insufficient institutional support for innovation (Dhir, 2021; Sherouk and Raad, 2020). Teachers in resource-constrained environments face particular challenges in incorporating technology-based or

collaborative learning strategies (Roshid and Haider, 2024). Moreover, social-emotional considerations, such as student motivation and engagement, remain crucial to successful implementation, especially in virtual and hybrid music learning contexts (Williams, 2021). These challenges highlight the urgent need for well-designed curriculum frameworks and professional development programs that explicitly embed 21st-century competencies in vocal education.

## Rationale and aim of the study

While individual studies have explored aspects of 21st-century skill integration in music and vocal education, there remains a lack of comprehensive synthesis regarding how these skills are defined, applied, and challenged within the field. The fragmented nature of existing research underscores the need for a systematic approach to mapping current practices and identifying pedagogical implications. Therefore, this study adopts scoping review methodology, following the framework proposed by Arksey and O'Malley (2005), to examine the integration of 21st-century skills in vocal pedagogy. This review aims to (a) chart the definitions and applications of 21st-century competencies in vocal teaching, (b) identify prevailing challenges and gaps, and (c) outline directions for future research and teacher education in preparing future-ready vocal practitioners.

## Methodology

This study adopted a scoping review design, a structured form of evidence synthesis used to map the breadth and nature of research on a particular topic and to identify conceptual boundaries and knowledge gaps (Munn et al., 2018; Peters et al., 2022; Pollock et al., 2024; Verdejo et al., 2021; Qiu and Gu, 2022). Scoping reviews are particularly useful in fields where literature is diverse, interdisciplinary, or underexplored (Peters et al., 2015; Westphal et al., 2021; Khalil et al., 2025). Because vocal pedagogy intersects music education, performance, and learning sciences, this approach was well suited to systematically map how 21st-century skills are conceptualized and applied within this context. The review followed the five-stage framework proposed by Arksey and O'Malley (2005) and refined by Levac et al. (2010). To strengthen methodological transparency, it was also aligned with the Joanna Briggs Institute (JBI) guidance (Munn et al., 2018; Peters et al., 2022; Khalil and Tricco, 2022; von Elm et al., 2019), which emphasizes the Population-Concept-Context (PCC) structure, iterative data charting, and clear reporting of review procedures.

### Stage 1: determining the research question

The guiding question was: What does the existing literature reveal about the integration of 21st-century skills in vocal music education and pedagogy? Following the PCC framework recommended by JBI, the population comprised vocal educators and learners; the concept focused on 21st-century skills such as creativity, collaboration, critical thinking, communication, and digital literacy; and the context included formal and non-formal vocal education settings. This

structure ensured conceptual focus and methodological consistency throughout the review.

### Stage 2: identifying relevant studies

Two databases were selected: Scopus and EBSCOhost (Education Research Complete). Scopus was chosen for its multidisciplinary scope, extensive citation tracking, and global coverage (Baas et al., 2020; Alryalat et al., 2019), while EBSCOhost was selected for its depth in education-focused and full-text resources (Libmann, 2007; Kirillova, 2017). Together, these databases provided complementary coverage balancing breadth and pedagogical specificity. The Boolean search string applied was:

*("vocal music education" OR "vocal teaching" OR "singing education") AND ("21st century skills" OR "21st century learning")*

Searches were limited to peer-reviewed, full-text English publications, with no date restriction. Table 1 presents detailed search strings and applied filters. The initial search yielded three documents from Scopus and seven from EBSCOhost. After removal of duplicates and one irrelevant record, nine studies were retained for full-text review.

### Stage 3: study selection and screening

Study selection proceeded in two stages: title/abstract screening and full-text review. Inclusion criteria were (i) focus on vocal or singing education, (ii) explicit discussion of 21st-century skills, and (iii) publication in peer-reviewed English sources. Conference abstracts, reviews, and editorials were excluded. Tables 2, 3 summarize the inclusion and exclusion criteria applied to each database. After screening, nine studies were included (Scopus = 3; EBSCOhost = 6). To ensure transparency, the identification, screening, eligibility, and inclusion process is illustrated in Figure 1, a flow diagram adapted from the PRISMA model (Moher et al., 2015) to enhance clarity in reporting, consistent with current recommendations for transparent scoping review procedures (Miller and Colquhoun, 2020).

### Stage 4: charting the data

Data extraction used a structured charting matrix that included author/year, publication type, study purpose, conceptualization of 21st-century skills, pedagogical approach, educational context, challenges, and key findings. This ensured comparability and reproducibility. The charting framework is shown in Table 4. Extraction was conducted manually and refined iteratively, following JBI principles of transparency and consistency (Hadie, 2024; Brodie et al., 2023).

TABLE 1 Search strings.

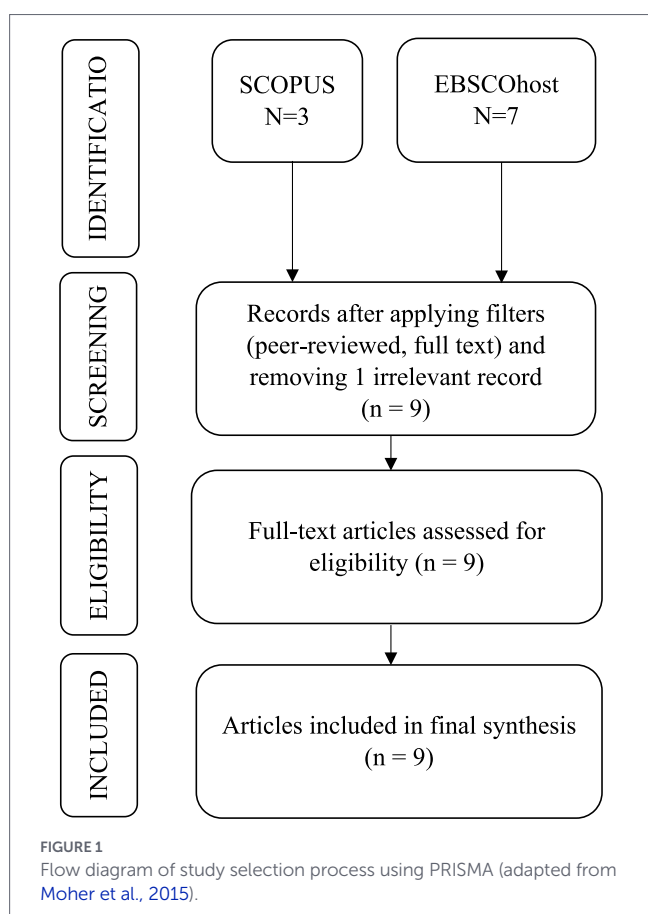
| Database  | Search string                                                                                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scopus    | <i>TITLE-ABS-KEY(("vocal music education" OR "vocal teaching" OR "singing education") AND ("21st century skills" OR "21st century competencies" OR "21st century learning" OR "future-ready skills"))</i> |
| EBSCOhost | <i>"vocal music education" OR "vocal teaching" OR "singing education" AND "21st century skills" OR "21st century learning"</i>                                                                            |

TABLE 2 Criteria for scopus.

| Criteria      | Inclusion             | Exclusion                              |
|---------------|-----------------------|----------------------------------------|
| Document type | Article, Book Chapter | Conference papers, reviews, editorials |
| Language      | English               | Non-English                            |

TABLE 3 Criteria for EBSCOhost.

| Criteria         | Inclusion                  | Exclusion                   |
|------------------|----------------------------|-----------------------------|
| Publication type | Academic Journal, Articles | Reports, descriptive papers |



## Stage 5: summarizing and reporting results

The charted data were analysed thematically to identify recurring patterns and relationships across the included studies. Three overarching themes emerged:

- The definitions and conceptualizations of 21st-century skills within vocal education
- Pedagogical strategies and applications in vocal instruction and
- Challenges and implications for teacher training and curriculum development.

These themes structure the Results section, guiding the interpretation of Table 4 and the synthesis of findings. The overall process of

evidence identification, selection, and analysis is illustrated in Figure 1, which depicts the workflow followed in this scoping review.

## Methodological scope and limitations

It is important to clarify that this review adopted a scoping rather than systematic review approach because the purpose was to map conceptual and pedagogical trends rather than to evaluate intervention effectiveness (Munn et al., 2018; Peters et al., 2015). Literature explicitly linking vocal pedagogy with 21st-century skills remains limited, with much singing pedagogy research focusing primarily on technical development rather than broader competency frameworks or educational outcomes (Vasil et al., 2019; Yoo, 2022). Consequently, studies were included only when they explicitly engaged with pedagogical, cognitive, technological, or socio-cultural competencies aligned with 21st-century learning. This conceptual filtering necessarily reduced the number of eligible studies but ensured thematic coherence. The review therefore does not claim to represent the entirety of vocal pedagogy scholarship but instead maps emerging intersections between vocal instruction and contemporary competency-based education, consistent with the mapping purpose of scoping reviews (Peters et al., 2015; Munn et al., 2018). Future reviews may broaden search terms or include additional databases to expand coverage. This limited number reflects the specificity of search criteria requiring explicit linkage between vocal pedagogy and 21st-century competencies rather than a lack of research in vocal pedagogy overall.

## Results

Following the inclusion and exclusion process detailed earlier, nine studies met the criteria for this scoping review. These studies, retrieved from the Scopus and EBSCOhost databases, encompass diverse educational, cultural, and methodological contexts, ranging from empirical investigations in formal music education to conceptual analyses of digital and reflective pedagogies. Collectively, they represent the growing scholarly effort to align vocal instruction with the competencies demanded in 21st-century education, including creativity, critical thinking, collaboration, communication, and technological literacy. The extracted data were systematically charted according to the review framework, with key information on study aims, conceptual definitions, pedagogical models, and outcomes synthesized in Table 4. This structured charting enabled both descriptive and interpretive analysis, revealing how each study conceptualizes and operationalizes 21st-century skills within vocal or singing education.

The nine included studies collectively demonstrate the increasing emphasis on reflective, student-centered, and technology-integrated approaches to vocal pedagogy. The conceptual focus of these studies reveals a consistent shift toward cognitive, creative, and social competencies aligned with global frameworks of 21st-century education. Across this body of work, the integration of 21st-century skills was achieved through diverse instructional models, including transformative, reflective, and AI-assisted pedagogies. Studies such as Wang et al. (2025) and İlkay et al. (2025) highlight that technology and collaboration serve as powerful mediators of learning, while Sweet and Parker (2019) emphasize reflective identity work as central to self-regulated skill development. Furthermore, the pedagogical diversity evident in Xu et al. (2025), Zhang (2016), and Zhang and Xing (2018) underscore

TABLE 4 Data charting matrix of included studies.

| Authors and year                        | Publication type | Database | Purpose/focus                                                                                         | Conceptualization of 21st-century skills                                                                            | Pedagogy approach                                            | Education context                                | Key findings/ implications                                                                                           | Study type/ participants                                                                                                        |
|-----------------------------------------|------------------|----------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Wang et al. (2025)</a>      | Journal Article  | Scopus   | Examines SVVR-based evaluation to enhance reflection and assessment in vocal music learning.          | Digital literacy, critical thinking, and reflective learning facilitated through immersive evaluation environments. | AI-assisted evaluation– explanation– comparison scaffolding. | Higher education vocal music courses.            | Technology-supported evaluator roles improved music appreciation, singing skills, motivation, and critical thinking. | Quasi-experimental study involving 55 university students (28 experimental; 27 control).                                        |
| <a href="#">İlkay et al. (2025)</a>     | Journal Article  | Scopus   | Investigates integration of 21st-century skills into ear training through novel classroom activities. | 4Cs framework extended with systematic thinking and technological literacy.                                         | Transformative, activity-based collaborative pedagogy.       | Undergraduate music teacher education.           | Activities enhanced engagement, collaboration, adaptability, and critical thinking in ear training contexts.         | Qualitative action research with 9 music education students, supported by evaluation from 15 field experts.                     |
| <a href="#">Weidner (2024)</a>          | Book Chapter     | Scopus   | Discusses disruption of traditional teaching cycles in music classrooms.                              | Innovation, adaptability, and reflective teaching practices.                                                        | Disruptive pedagogy promoting reflective curriculum reform.  | Music education classroom contexts.              | Pedagogical disruption supports reflective teaching practices and curricular innovation.                             | Conceptual pedagogical analysis; participant details not reported.                                                              |
| <a href="#">Xu et al. (2025)</a>        | Journal Article  | EBSCO    | Explores adaptation of Napoli singing techniques in contemporary vocal instruction.                   | Integration of tradition, adaptability, and intercultural competence.                                               | Comparative and integrative cross-cultural pedagogy.         | University vocal programs.                       | Hybrid pedagogical approaches strengthen vocal technique and intercultural understanding.                            | Qualitative pedagogical development study with expert evaluations; participant details not reported.                            |
| <a href="#">Sweet and Parker (2019)</a> | Journal Article  | EBSCO    | Examines female vocal identity development through lived experiences.                                 | Metacognition, self-awareness, and reflective learning.                                                             | Reflective and identity-based pedagogy.                      | Secondary and tertiary vocal education contexts. | Vocal identity development influences future pedagogical and performance trajectories.                               | Phenomenological study involving 39 university vocal students aged 19–35.                                                       |
| <a href="#">Zhang and Xing (2018)</a>   | Journal Article  | EBSCO    | Investigates pronunciation training within college vocal instruction.                                 | Communication skills and cognitive adaptability.                                                                    | Integrated pronunciation and vocal training pedagogy.        | Undergraduate music education programs.          | Integrated language and vocal instruction improved communication and performance expression.                         | Empirical study using survey data and classroom intervention; participant details not reported.                                 |
| <a href="#">Zhang (2016)</a>            | Journal Article  | EBSCO    | Examines integration of Western and Chinese vocal traditions in pedagogy.                             | Cultural literacy and adaptive creativity.                                                                          | Cross-cultural hybrid vocal pedagogy.                        | Higher music education institutions.             | Fusion pedagogy supports development of culturally versatile singers.                                                | Pedagogical and historical teaching-method analysis based on institutional observations; no formal participant sample reported. |

(Continued)



TABLE 4 (Continued)

| Authors and year         | Publication type | Database | Purpose/focus                                                             | Conceptualization of 21st-century skills                                  | Pedagogy approach                                  | Education context                                  | Key findings/implications                                                                                           | Study type/ participants                                                 |
|--------------------------|------------------|----------|---------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Baughman (2016)          | Journal Article  | EBSCO    | Explores strategies for improving practice habits in voice instruction.   | Self-regulation, goal-setting, and reflective practice.                   | Deliberate practice and metacognitive instruction. | Private and institutional vocal teaching contexts. | Structured reflection enhances independent practice and lifelong learning skills.                                   | Practice-based pedagogical discussion; participant details not reported. |
| Collins and Wells (2014) | Journal Article  | EBSCO    | Reimagines ensemble teaching using student-centered collaborative models. | Collaboration, creativity, and problem-solving through ensemble learning. | Project-based and collaborative ensemble pedagogy. | Secondary and collegiate ensemble settings.        | Student-centered ensemble practices promote collaboration, creativity, learner agency, and inclusive participation. | Professional practice report; participant details not reported.          |

that 21st-century competencies in vocal education extend beyond performance excellence to encompass cultural awareness, communication, and adaptability. Collectively, these findings suggest that effective vocal instruction in the modern era must balance technical artistry with critical, reflective, and intercultural dimensions of learning. Beyond cognitive and technological competencies, several scholars emphasize the centrality of embodiment, desire, and personal agency in vocal learning. Rocha (2024), for instance, conceptualizes vocal pedagogy as an embodied and affective practice that foregrounds bodily awareness, individuality, and authorial voice, aligning closely with the emotional-social and reflective dimensions identified in this review.

This synthesis is illustrated in Figure 2, which conceptualizes the integration of 21st-century skills within vocal pedagogy as an interconnected framework involving cognitive, creative, technological, emotional-social, and cultural dimensions. Complementary literature beyond the nine included studies highlights adjacent pedagogical developments. First, reviews of modern learning and microlearning indicate a growing body of evidence that bite-sized learning (short, scaffolded micro-units distributed via platforms such as TikTok and mobile apps) can support personalization, reduce cognitive load, and increase student engagement in music learning contexts (Mazlan et al., 2023; Mazlan et al., 2025a, b). Second, a recent scoping review focused on vocal health in China (Xiao and Mazlan, 2024) signals that technological and pedagogical change must be accompanied by curricular attention to anatomy, hygiene, and injury prevention to avoid unintended health consequences for singers, particularly in online or compressed-practice environments. Third, discipline-specific innovation (e.g., jazz reharmonization applied to Malay Asli music) illustrates how cross-cultural hybridization can function as a pedagogical strategy for creative skill-building and decolonizing curricular norms (Mazlan et al., 2024). These complementary sources strengthen the review’s position that technological affordances (including microlearning) can foster 21st-century competencies in vocal pedagogy, but that success depends on reflective practice, attention to health, and culturally responsive design.

Figure 2 illustrates the multidimensional relationship between 21st-century competencies and vocal pedagogy. At the core lies the *Vocal Pedagogy Process*, surrounded by five interrelated domains that represent how these competencies manifest in vocal education:

1. Cognitive Dimension – Critical thinking, reflection, and metacognitive awareness (Baughman, 2016; Sweet and Parker, 2019).
2. Creative Dimension – Innovation, improvisation, and problem-solving within vocal expression (İlkay et al., 2025; Weidner, 2024).
3. Technological Dimension – Digital literacy, AI-assisted learning, and technology-mediated collaboration (Wang et al., 2025).
4. Emotional-Social Dimension – Empathy, expressive awareness, and collaborative musicianship developed through ensemble and identity-based learning (Sweet and Parker, 2019; Collins and Wells, 2014).
5. Cultural Dimension – Global awareness and intercultural adaptability through cross-genre and cross-tradition learning (Zhang, 2016; Xu et al., 2025).

These domains intersect to form a holistic pedagogical model, positioning 21st-century skills as enablers of reflective and adaptive

## Integration of 21st-Century Skills in Vocal Pedagogy

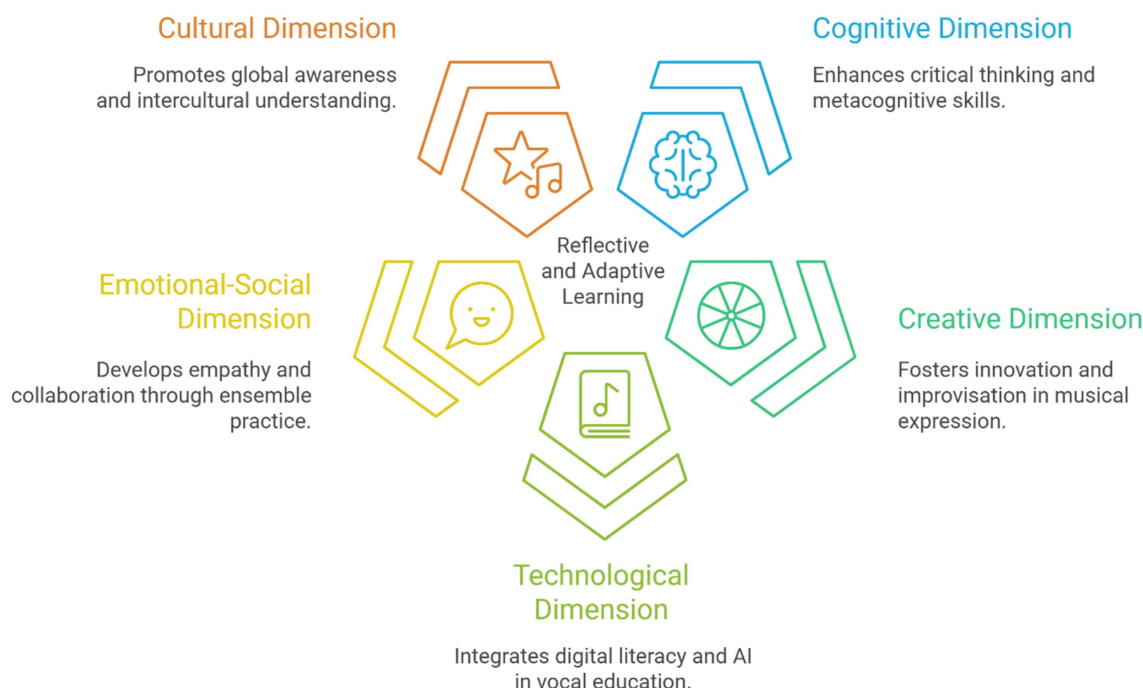


FIGURE 2

Conceptual framework of 21st-century skill integration in vocal pedagogy. The framework positions *reflective and adaptive learning* at the centre of the vocal pedagogy process, surrounded by five interrelated domains representing 21st-century competencies: cognitive, creative, technological, emotional–social, and cultural. The model illustrates how each dimension supports a holistic approach to vocal education, integrating critical, creative, digital, emotional, and intercultural skills into reflective teaching and learning practices (Source: Authors).

learning rather than as discrete competencies. The framework underscores that contemporary vocal education must integrate these domains to cultivate agile, interculturally aware, and self-regulated learners capable of navigating complex musical and professional environments. The AI dimension in music education complements findings from this review by showcasing how algorithmic tools can augment student reflection and self-regulated learning. Mazlan et al. (2025a, b) identified that AI-based systems, particularly in personalized vocal coaching and performance analytics, enhance both efficiency and feedback precision. This aligns with the 21st-century skill domains identified in Figure 2, especially within the technological and cognitive dimensions, reinforcing that digital and reflective competencies develop synergistically when AI is integrated purposefully into vocal pedagogy.

## Discussion

The findings of this scoping review reveal emerging tendencies toward transformation in vocal pedagogy from tradition-bound transmission models toward reflective, technology-enhanced, and student-centered practices. This transition echoes the global educational discourse on 21st-century skills, where adaptability, collaboration, and creativity are increasingly viewed as core learning outcomes (Bhatnagar, 2020; Yoo, 2022; Naixue et al., 2024). Within vocal education, this shift is not merely procedural but

epistemological, it redefines how musical knowledge, artistic identity, and learning agency are constructed and enacted. Recent scholarship in vocal and instrumental pedagogy confirms this epistemic shift, documenting a move from transfer-oriented instruction toward transformative, learner-centred, and reflective models of one-to-one teaching (Carey et al., 2021; Cox and Forbes, 2022; Almqvist and Werner, 2024), with parallel discussions highlighting the evolving role of pedagogical practices in shaping 21st-century competencies (Gill, 2021; Zhang and Xing, 2018). These studies demonstrate that contemporary vocal pedagogy increasingly positions teachers not as authoritative transmitters of knowledge but as reflective facilitators who co-construct learning trajectories with students.

## Redefining the pedagogical core: from technique to thinking

A critical synthesis of the reviewed studies suggests that 21st-century skill integration moves vocal teaching beyond technical mastery toward higher-order cognition and self-regulated artistry. Baughman (2016) and Sweet and Parker (2019) demonstrate that metacognition and self-reflective identity formation constitute central pillars of contemporary vocal education. These findings challenge the enduring master–apprentice paradigm, which privileges imitation and conformity over reflection and innovation. Instead, reflective practice reframes technique as an outcome of cognitive and emotional awareness rather than mere muscular control. This reconceptualization

situates vocal pedagogy within the broader cognitive apprenticeship model, where learners develop through guided reflection and dialogic interaction, a perspective long underrepresented in music education research.

## Technological mediation and pedagogical tensions

The role of technology, particularly as explored by Wang et al. (2025), introduces both opportunities and tensions in redefining the teacher–learner dynamic. AI-assisted feedback systems enhance precision and accessibility but risk diminishing embodiment, aesthetic, and relational dimensions of voice teaching. While technological mediation fosters efficiency and inclusivity, its overemphasis may inadvertently promote mechanistic learning, potentially reducing the voice to quantifiable data. This antithesis between *efficiency* and *expressivity* underscores the necessity of pedagogical balance, where digital tools augment, rather than replace, the humanistic essence of vocal pedagogy. Such a stance aligns with Weidner's (2024) “disruptive pedagogy,” which calls for critical engagement with technology as a medium of reflection, not dependency. The intersection of AI and vocal pedagogy reveals both opportunities and tensions. As Mazlan et al. (2025a, b) demonstrate, AI-assisted tools have begun reshaping assessment, practice, and feedback cycles in music learning, fostering self-regulation and personalization. Yet, uncritical adoption risks reducing musical expression to quantifiable metrics, echoing concerns raised by earlier literature on the dehumanization of arts education. Thus, AI should be treated not as a replacement for teacher artistry but as a *reflective partner*, a means to extend teachers' diagnostic insight while preserving the aesthetic, embodied, and cultural essence of singing. Critically, this balance mirrors the methodological ethos of your scoping review: technology integration must be reflective, context-aware, and underpinned by humanistic pedagogy.

## Cultural hybridity and intercultural agency

Another central insight emerging from the synthesis is the reconceptualization of “vocal artistry” as a culturally hybrid and globally adaptive construct. The works of Xu et al. (2025) and Zhang (2016) exemplify how integrating diverse vocal traditions cultivates intercultural literacy, a core 21st-century competency that transcends technical skill. This approach contrasts sharply with traditional conservatoire curricula, which often privilege Western classical norms. The infusion of cross-cultural perspectives signals a necessary pedagogical decolonization, repositioning the learner as a global citizen-artist. However, this hybridization also demands new frameworks for assessing quality and authenticity, raising epistemological questions about how “excellence” is defined within multicultural vocal spaces.

The case study on jazz reharmonization for Malay Asli music (Mazlan et al., 2024) exposes how *decolonial* and creative pedagogies can expand the musical canon and cultivate versatile expressive skills. However, such hybridizations also raise an epistemological question: *how do educators evaluate authenticity and artistic integrity while nurturing creative hybridity?* The answer suggested by the case study and by broader scoping findings is that curricular frameworks must foreground *contextualized assessment criteria*, reflexive pedagogy, and student agency. In practical terms, this means assessment rubrics that

value intercultural fluency, imaginative adaptation, and ethical appropriation practices alongside technical metrics.

## Vocal health as an ethical and curricular imperative

Xiao and Mazlan's (2024) scoping review foregrounds an often-overlooked dimension of modern vocal pedagogy: vocal health. Technological convenience and compressed practice modalities may inadvertently elevate the risk of voice strain, particularly where students lack direct instruction in hygiene and anatomically informed practice. Integrating vocal health into 21st-century pedagogies is therefore not optional; it is a curricular and ethical requirement. This implies teacher education must include vocal-health literacy and that microlearning packages should embed short but explicit modules on warm-up, vocal rest, and risk mitigation, assuring that digital innovation does not come at the expense of learner wellbeing. Bringing these threads together produces a guardedly optimistic conclusion: microlearning and AI-enhanced tools can accelerate the adoption of 21st-century skills in vocal pedagogy, but they must be coupled with (1) reflective, scaffolded learning cycles; (2) explicit vocal-health instruction; and (3) culturally responsive curriculum design. Without these conditions, technological adoption risks being superficial or even harmful; with them, it can meaningfully transform vocal training toward adaptive, creative, and health-aware practice.

## Microlearning and the promise/limits of short-form pedagogy

The inclusion of bite-sized learning literature complicates the promise of technology in vocal pedagogy. On the one hand, microlearning promises accessibility, personalization, and higher engagement, qualities that align tightly with the 21st-century skill agenda and the student-centered approaches identified across the nine primary studies (Mazlan et al., 2023). Short units can scaffold practice and immediate feedback loops, and fit modern learners' schedules, making them especially compatible with blended and flipped approaches. On the other hand, the microlearning model risks truncating embodied practice: singing requires sustained motor learning, breath control, and holistic practice sessions that may not map neatly onto bite-sized chunks without careful design. This creates a pedagogical tension: microlearning can support 21st-century competencies (e.g., digital literacy, self-regulation) while simultaneously demanding deliberate scaffolding to preserve embodiment, vocal stamina, and artistic development. Practically, this suggests blended designs where short-form modules are embedded within longer, reflective practice cycles rather than replacing them (Mazlan et al., 2023; Mazlan et al., 2025b). This perspective is further supported by emerging discussions on AI-integrated pedagogical approaches in music education (Mazlan et al., 2025a).

## The reflective-transformative nexus

Ilkay et al. (2025) and Collins and Wells (2014) illustrate how collaborative and ensemble-based pedagogies foster social and emotional intelligence, competencies central to lifelong learning. These findings reveal a nexus between reflective and transformative



teaching: reflection enables personal insight, while transformation actualizes that insight through social interaction and creative agency. This interplay resonates with constructivist principles, positioning the voice not only as an instrument of sound but as a site of identity, empathy, and social connection. Importantly, such models demonstrate that 21st-century skills are not external additions to vocal curricula but intrinsic qualities of a relational, reflective learning ecology.

## Pedagogical gaps and systemic barriers

Despite this conceptual progression, the review exposes persistent systemic gaps. Many studies highlight the limited institutional and professional structures supporting educators' capacity to enact these paradigms. Curriculum rigidity, insufficient digital training, and lack of reflective teacher education (Nazarian et al., 2024; Pharamela and Singh-Pillay, 2025) continue to hinder meaningful integration of 21st-century skills. This discrepancy between policy and practice mirrors findings across broader education research, suggesting that transformative pedagogical change demands not only new content but also new epistemologies of teaching, those that value process, reflection, and cultural multiplicity as equally as performance outcomes. These findings carry particular implications for music teacher education, where future educators must be equipped not only with technical vocal expertise but also with pedagogical strategies that foster creativity, collaboration, digital literacy, and intercultural awareness among learners. However, given that only nine studies met inclusion criteria, findings should be interpreted cautiously and cannot be assumed to represent global vocal pedagogy practices. Instead, they highlight emerging tendencies within selected educational contexts.

## Conclusion

This scoping review critically maps how 21st-century skills are conceptualized and applied in vocal pedagogy, with particular relevance for music teacher education and curriculum development. The synthesis underscores that modern vocal education is no longer defined by mastery of technique alone but by the cultivation of adaptive, critically engaged, and culturally literate practitioners. The review contributes three key insights. First, vocal pedagogy is undergoing an epistemic shift from a teacher-centered, skill-reproduction paradigm to a learner-centered, meaning-construction paradigm, where reflection and creativity drive mastery. Second, technology serves as both an enabler and disruptor, amplifying accessibility and feedback precision while necessitating vigilance against dehumanization of the artistic process. Third, intercultural integration redefines excellence, positioning cultural adaptability as a marker of artistic and pedagogical maturity. Nevertheless, the study identifies structural limitations, notably the lack of explicit frameworks connecting 21st-century competencies to assessment and teacher training. Addressing these gaps requires curriculum reform that embeds reflection, collaboration, and digital fluency as core learning outcomes, not ancillary skills.

Future research should examine longitudinal impacts of reflective and AI-mediated pedagogies on vocal development, as well as

explore policy-level strategies for embedding 21st-century frameworks in music teacher education. Future studies may also examine conference proceedings, professional voice associations, and practitioner-oriented publications to determine how emerging pedagogical trends are reflected in professional communities beyond academic journals. In addition to reflective and technology-mediated practices, future vocal pedagogy must explicitly design micro-learning sequences that preserve embodied practice and embed vocal-health literacy; curriculum reform should equally prioritize intercultural competence as a marker of modern musical excellence. Calls for reflective, student-centred, and evidence-based vocal pedagogy are not new but are increasingly urgent. Historical analyses of vocal pedagogy illustrate a gradual shift from prescriptive treatise-based instruction toward interdisciplinary, scientifically informed, and learner-responsive approaches (Viales-Montero, 2022). Contemporary voice pedagogy research further argues that reflective awareness, problem-solving, and metacognitive engagement are essential for developing adaptable and resilient vocal practitioners in 21st-century contexts (Lã, 2017).

Ultimately, cultivating 21st-century vocal practitioners demands a philosophical reorientation beyond methodological innovation. Vocal pedagogy must embrace a holistic, human-centered model where technique, cognition, emotion, and culture converge, producing not just skilled performers, but adaptive, reflective, and empathetic artists equipped for the complexities of the contemporary world. Collectively, these studies suggest that the integration of 21st-century skills in vocal pedagogy is less a matter of introducing new tools than of re-examining long-standing pedagogical assumptions about authority, embodiment, authenticity, and learner agency.

## Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding authors.

## Author contributions

WL: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. CANM: Conceptualization, Project administration, Supervision, Validation, Writing – original draft, Writing – review & editing. Data curation, Methodology, Software. ARS: Conceptualization, Formal analysis, Methodology, Supervision, Validation, Visualization, Writing – review & editing.

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## Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

## Generative AI statement

The author(s) declared that Generative AI was used in the creation of this manuscript. Generative AI was used in the preparation of this manuscript for language refinement and clarity improvement only. All intellectual content, data analysis, interpretation, and conclusions were developed by the authors. The authors have reviewed and verified the accuracy of all AI-assisted text and take full responsibility for the final content.

## References

- Alharbi, N. S. (2024). Exploring the perspectives of cross-cultural instructors on integrating 21st century skills into EFL university courses. *Front. Educ.* 9:1302608, 1–12. doi: 10.3389/feduc.2024.1302608
- Almqvist, C. F., and Werner, A. (2024). Maintaining and challenging conservative teaching and learning culture in conservatories: the need for holistic pedagogy in educational fields of tension. *Res. Stud. Music Educ.* 46, 257–270. doi: 10.1177/1321103X231187766
- Alryalat, S. A. S., Malkawi, L. W., and Momani, S. M. (2019). Comparing bibliometric analysis using PubMed, Scopus, and web of science databases. *J. Vis. Exp.* 152:e59099. doi: 10.3791/59099
- Arksey, H., and O'Malley, L. (2005). Scoping studies: towards a methodological framework. *Int. J. Soc. Res. Methodol.* 8, 19–32. doi: 10.1080/1364557032000119616
- Baas, J., Schotten, M., Plume, A., Côté, G., and Karimi, R. (2020). Scopus as a curated, high-quality bibliometric data source for academic research in quantitative science studies. *Quant. Sci. Stud.* 1, 377–386. doi: 10.1162/qss\_a\_00019
- Baughman, M. (2016). Incorporating practice strategies into voice instruction. *J. Sing.* 72, 215–228. Available at: <https://www.proquest.com/scholarly-journals/incorporating-practice-strategies-into-voice/docview/1780076236/se-2>
- Bhatnagar, A. (2020). Assessment of 21st century skills in STEM programs. *J. Eng. Educ. Transform.* 33, 556–560. doi: 10.16920/jeet/2020/v33i0/150116
- Brodie, T., Howard, N. J., Pearson, O., and Brown, A. (2023). Enhancement of scoping review methodology to reflect aboriginal and Torres Strait islander ways of knowing, being and doing. *Aust. N. Z. J. Public Health* 47, 506–512. doi: 10.1016/j.anzjph.2023.100096
- Carey, G., Grant, C., and Cifuentes, M. V. (2021). Teachers of instruments, or teachers as instruments? From transfer to transformative approaches to one-to-one pedagogy. *Rev. Mus. Chil.* 75, 175–183. doi: 10.4067/S0716-27902021000200175
- Colahan, E. (2022). Conducting a laboratory pedagogy: empowering the music apprentice as a knowledge producer in the classroom. *Coll. Music. Symp.* 62, 22–52. doi: 10.18177/sym.2022.62.sr.11559
- Collins, C. L., and Wells, J. (2014). Professional notes: 21st-century ensembles—what we imagine, we can become! *Music. Educ. J.* 100, 18–21. doi: 10.1177/0027432114521994
- Cox, D., and Forbes, M. (2022). Introducing multi-sited focused ethnography for researching one-to-one (singing voice) pedagogy in higher education. *Music. Educ. Res.* 24, 625–637. doi: 10.1080/14613808.2022.2138842
- Dhir, H. K. (Ed.) (2021). Handbook of research on barriers for teaching 21st-century competencies and the impact of digitalization. Hershey, PA: IGI Global.
- Gibson, C. L. (2026). The master and the apprentice: educational power and musical culture in the “wild, wild west”. *Res. Stud. Music Educ.* 1–17. doi: 10.1177/1321103X251404287
- Gill, P. (2021). A concise, user-friendly guide to conducting and reporting scoping reviews in healthcare. *Pac. Rim Int. J. Nurs. Res.* 25, 421–430. Available online at: <https://he02.tci-thaijo.org/index.php/PRIJNR/article/view/251663>
- Hadie, S. N. H. (2024). ABC of a scoping review: a simplified JBI scoping review guideline. *Educ. Med. J.* 16, 35–45. doi: 10.21315/eimj2024.16.2.14
- İlkay, G., Kıvanç Öztuğ, E., Eren, H. C., and Sulun, N. (2025). Transformative 21st century approaches in musical ear training: fostering essential skills for enhanced learning. *SAGE Open* 15, 1–15. doi: 10.1177/2158244025132643
- Irfiana, K., Purnawarman, P., and Sukyadi, D. (2023). Exploring EFL teachers' assessment for 21st century skills: practices and challenges. *AIP conference proceedings*.
- Khalil, H., Jia, R., Moraes, E. B., and Evans, C. (2025). Scoping reviews and their role in identifying research priorities. *J. Clin. Epidemiol.* 167, 50–59. doi: 10.1016/j.jclinepi.2025.111712
- Khalil, H., and Tricco, A. C. (2022). Differentiating between mapping reviews and scoping reviews in the evidence synthesis ecosystem. *J. Clin. Epidemiol.* 149, 175–182. doi: 10.1016/j.jclinepi.2022.05.012
- Kirillova, O. V. (2017). Scopus database as a source of representing Bulgarian science to the international academic community. *Digit. Preserv. Pres. Cult. Sci. Herit.* 7, 35–42. Available online at: <https://dipp.math.bas.bg/dipp/article/download/dipp.2017.7.6/pdf>
- Lâ, F. M. B. (2017). Towards evidence-based practices in voice pedagogy. *Rev. Logop. Foniatr. Audiol.* 37, 180–187. doi: 10.1016/j.rlfa.2017.07.002
- Laato, S., Rauti, S., and Sutinen, E. (2020). The role of music in 21st century education—comparing programming and music composing. *Proceedings of the IEEE 20th international conference on advanced learning technologies (ICALT 2020)*.
- Levac, D., Colquhoun, H., and O'Brien, K. K. (2010). Scoping studies: advancing the methodology. *Implement. Sci.* 5, 1–9. doi: 10.1186/1748-5908-5-69
- Libmann, F. (2007). Web of science, Scopus, and classical online: philosophies of searching. 31, 16–21. Available online at: <https://pascal-francis.inist.fr/vibad/index.php?action=getRecordDetail&idt=18861431> (Accessed December 15, 2025).
- Long, M., Creech, A., Gaunt, H., and Hallam, S. (2014). Conservatoire students' experiences and perceptions of instrument-specific master classes. *Music. Educ. Res.* 16, 176–192. doi: 10.1080/14613808.2013.859659
- Mazlan, C. A. N., Abdullah, M. H., Abdullah, M. A., Hidayatullah, R., Jamnongsarn, S., and Hassim, D. S. H. A. (2024). Jazz guitar reharmonization in Malay Asli music: a case study on harmonic adaptation in pedagogy. *Resital: Jurnal Seni Pertunjukan* 25, 460–516. doi: 10.24821/resital.v25i3.14907
- Mazlan, C. A. N., Abdullah, M. H., Sulong, M. A., Abas, A., Ramdan, M. R., Safian, A. R., et al. (2023). Exploring the integration of bite-sized learning: a scoping review of research in education and related disciplines. *Contemp. Educ. Technol.* 15:ep468. doi: 10.30935/cedtech/13622
- Mazlan, C. A. N., Koning, S. I., Yusoff, M. Y. M., Awang Hassim, D. S. H., Hidayatullah, R., and Jamnongsarn, S. (2025a). Artificial intelligence in music education: trends, challenges, and future directions. In *Proceedings of the 2025 5th international conference on artificial intelligence and education (ICAIE)* (pp. 1–8). IEEE
- Mazlan, C. A. N., Yusoff, M. Y., Safian, A. R., Koning, S. I., Taib bin Saearani, M. F., Md Noor, A. R., et al. (2025b). The evolution of modern learning: a dual systematic review of emerging trends and challenges. *Asian J. Univ. Educ.* 21, 793–813. doi: 10.24191/ajue.v21i3.54
- Miller, E., and Colquhoun, H. (2020). The importance and value of reporting guidance for scoping reviews: a rehabilitation science example. *Aust. J. Adv. Nurs.* 37, 15–23. doi: 10.37464/2020.37.148
- Moher, D., Liberati, A., Tetzlaff, J., and Altman, D. G. the PRISMA Group (2015). Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Syst. Rev.* 4, 1–9. doi: 10.1016/j.jisu.2010.02.007
- Moor, A. (2018). The authentic actor: an examination of vocal training within the Australian conservatory model. *Voice Speech Rev.* 12, 256–266. doi: 10.1080/23268263.2018.1466763

- Munn, Z., Peters, M. D. J., Stern, C., Tufanaru, C., McArthur, A., and Aromataris, E. (2018). Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. *BMC Med. Res. Methodol.* 18:143. doi: 10.1186/s12874-018-0611-x
- Naixue, B., Chang, Y.-L., and Wing, C. K. (2024). The differences in 21st century 4C skills among diverse university choirs in China. *Malaysian J. Music.* 13, 1–17. doi: 10.37134/mjm.vol13.1.1.2024
- Nazarian, M., Alsheikh, N., and Alhosani, M. (2024). Between vision and revision: English language teachers accentuating their voices about 21st century skills. *Int. J. Learn. Higher Educ.* 31, 25–50. doi: 10.18848/2327-7955/CGP/v31i01/25-50
- O'Bryan, J., and Harrison, S. (2024). Musical theatre education and training in the 21st century. New York, NY: Routledge. doi: 10.4324/9780429504440
- Peters, M. D. J., Godfrey, C. M., Khalil, H., McInerney, P., Parker, D., and Soares, C. B. (2015). Guidance for conducting systematic scoping reviews. *Int. J. Evid. Based Healthc.* 13, 141–146. doi: 10.1097/XEB.0000000000000050
- Peters, M. D. J., Godfrey, C. M., McInerney, P., Munn, Z., Tricco, A. C., and Khalil, H. (2022). Best practice guidance and reporting items for the development of scoping review protocols. *JBIM Evidence Synthesis* 20, 953–968. doi: 10.111124/JBIES-21-00242
- Pharamela, S., and Singh-Pillay, A. (2025). Technical and vocational education and training college lecturers and 21st-century skills: awareness and implications for teaching practices. *J. Technic. Educ. Train.* 17, 182–196. doi: 10.30880/jtet.2025.17.02.013
- Pollock, D., Evans, C., Menghao Jia, R., and Munn, Z. (2024). “How-to”: scoping review? *J. Clin. Epidemiol.* 162, 111–120. doi: 10.1016/j.jclinepi.2024.111572
- Qiu, R., and Gu, Y. (2022). Interpretation of the PRISMA extension for scoping review (PRISMA-ScR). *Chin. J. Evid.-Based Med.* 22, 706–712. doi: 10.7507/1672-2531.202203042
- Rocha, V. C. M. (2024). Perspectives for a vocal pedagogy: possibilities for a desiring voice. *Urdimento Journal of Studies in Performing Arts.* 3, 1–16. doi: 10.5965/1414573103522024e0203
- Roshid, M. M., and Haider, M. Z. (2024). Teaching 21st-century skills in rural secondary schools: from theory to practice. *Heliyon.* 10, 1–19. doi: 10.1016/j.heliyon.2024.e30769
- Sherouk, K., and Raad, K. (2020). What educational actions are urgently needed for developing e-learning and enhancing the 21st century skills? *Proceedings of the international conference on e-learning (ICEL).*
- Sweet, B., and Parker, E. C. (2019). Female vocal identity development: a phenomenological study. *J. Res. Music. Educ.* 67, 82–99. doi: 10.1177/0022429418809981
- Vasil, M., Weiss, L., and Powell, B. (2019). Popular music pedagogies: an approach to teaching 21st-century skills. *J. Music. Teach. Educ.* 28, 20–34. doi: 10.1177/1057083718817413
- Verdejo, C., Tapia-Benavente, L., Schuller-Martínez, B., and Silva-Dreyer, A. M. (2021). What you need to know about scoping reviews. *Medwave* 21:e8482. doi: 10.5867/medwave.2021.02.8144
- Viales-Montero, R. (2022). Historical overview of classical vocal pedagogy: a journey from the singing treatises to the voice sciences. *Rev. Electrón. Educare* 26, 506–525. doi: 10.15359/ree.26-2.27
- von Elm, E., Schreiber, G., and Haupt, C. C. (2019). Methodological guide for scoping reviews (JBI methodology). *Z. Evid. Fortbild. Qual. Gesundheitswes.* 141, 82–89. doi: 10.1016/j.zeq.2019.05.004
- Wang, H. J., Chen, K., Wang, Y., and Liu, C. C. (2025). From the performer to the evaluator: exploring the effects of the evaluation-explanation-comparison-based SVVR on students' vocal music appreciation and singing skills. *J. Comput. Assist. Learn.* 41:e70145. doi: 10.1111/jcal.70145
- Weidner, B. N. (2024). “Disruptive practices in music pedagogy,” in eds. M. Haning, J. A. Stevens, and B. N. Weidner. *Points of disruption in the music education curriculum*, vol. 2 Individual changes (New York, NY: Routledge). doi: 10.4324/9781003410669-1
- Westphal, K. K., Regoeci, W., Masotya, M., Vazquez-Westphal, B., Lounsbury, K., McDavid, L., et al. (2021). From Arksey and O'Malley and beyond: customizations to enhance a team-based mixed approach to scoping review methodology. *MethodsX* 8:101375. doi: 10.1016/j.mex.2021.101375
- Williams, R. (2021). “Innovative instructional methods integrating 21st-century competencies,” in ed. H. K. Dhir. *Handbook of research on barriers for teaching 21st-century competencies and the impact of digitalization*. Hershey, PA: IGI Global. 272–292. doi: 10.4018/978-1-7998-6967-2.ch015
- Wu, H. (2025). Modern Chinese and European vocalist training methods: an exploration of the involvement of bachelors with the help of IT sources. *Eur. J. Educ.* 60:e12847. doi: 10.1111/ejed.12847
- Xiao, G., and Mazlan, C. A. N. (2024). Vocal health in China: a scoping review of literature (2020–2024) utilizing abstract-targeted search strategies. *South East. Eur. J. Public Health XXV*, 2202–2220. doi: 10.70135/seejph.vi.3091
- Xu, X., Khosiditsayan, N., and Chalermchokchaisit, K. (2025). Singing techniques of Italian Napoli song for modern music education. *Int. Educ. Stud.* 18, 185–195. doi: 10.5539/ies.v18n5p185
- Yoo, H. (2022). Building 21st-century skills through technology in general music classes. *J. Gen. Music Educ.* 36, 21–31. doi: 10.1177/27527646221110867
- Zhang, W. (2016). An infusion of western and eastern singing styles into Chinese vocal art: introducing the teaching methodology for training Chinese bel canto singers. *J. Sing.* 72, 291–298. Available at: <https://www.proquest.com/scholarly-journals/infusion-western-eastern-singing-styles-into/docview/1758618546/se-2>.
- Zhang, Y., and Xing, J. (2018). Study on the application of pronunciation training in vocal music teaching in colleges and universities. *Educ. Sci. Theory Pract.* 18, 2054–2063. doi: 10.12738/estp.2018.5.104